loop constructs

- while
- do ... while
- for

Syntax:

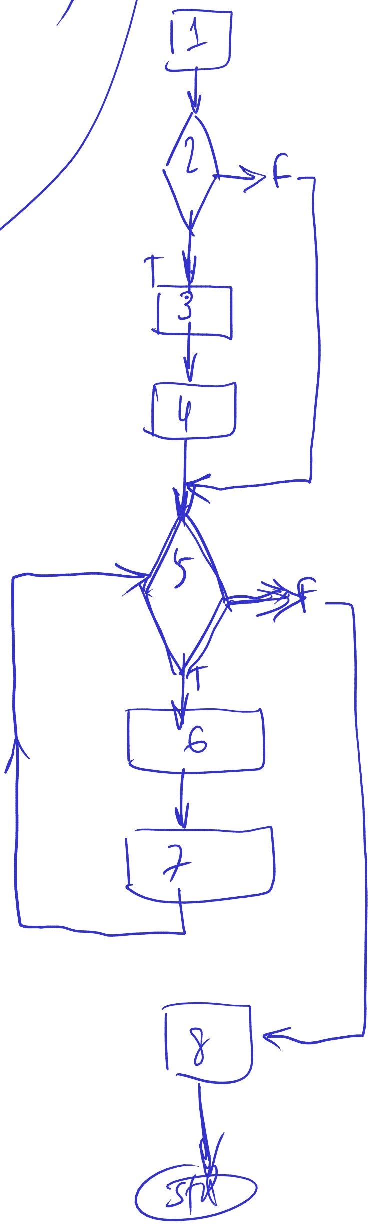
```
while (booleantest-cond)
{
  =
  Body of loop
  =
}
```

```
if (boolean test cond)
{
  =
  If Block;
  =
}
```

```
main(){ 1) int a=1, b=1;
        2) if (a<5) {
              3) pf("a=%d\n", a);
              4) ++a;
           }
        5 while (b<5) {
              6 pf("b=%d\n", b);
              7 ++b;
           }
        8 -
}
```

a = ~~1~~ 2

b = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

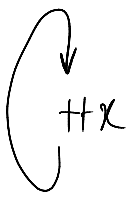


```
main() { int a=1;
   while(a)
    {
     pf("In loop \n");
    ++a;
    }
  Pf("after loop a=%d\n", a) ✓
 }
```

a => 1 - 2 - 3 - - - - - - - - 2.1bn => -2bn ····· -1bn ··· -1,0

Entry controlled

char x=1;
2
3
4
:
127
↓
-128
-127
⋮
-1
0



```
main() { int a=1;
   while(a);   →   while(a) {
    {                        }
     pf("In loop \n");
    ++a;
    }
  Pf("after loop a=%d\n", a);
 }
```

gnfinite loop

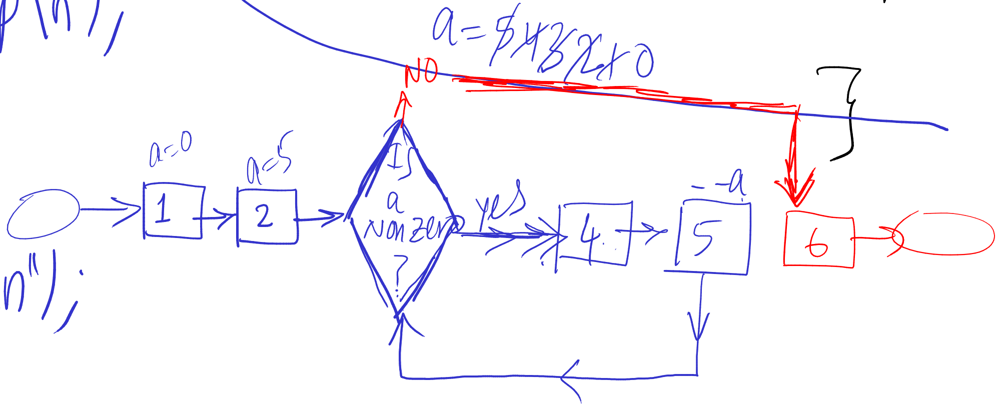
```
{
 Pf( —
 ++a;
}
```

```
main() {
 int a = 1;
 if(a)
   Pf("In if \n");
   ++a;
 else
   Pf("In else \n");
   a+=10;
 Pf("%d", a);
}
```

Error

Misplaced else

```
1) int a=0;
2) a=5;
3) while(a){
    4) Pf("In loop \n");
   5) --a;
   }
6) Printf("after loop \n");
}
```



Count the no-of-bits-set in an integer.

bit => 31, 30, 29 .... 1, 0 | -1

```
main() { int bit, cnt;
         int data;
       pf("Enter int:");
       sf("%d", &data);
bit=31; cnt=0;
while(bit>=0)
 {
   if((data>>bit)&1)        cnt++;

 --bit;
 }
```

```
TEST:  if(bit>=0)
       {
          if((data>>bit)&1)
                          cnt++;
          --bit;
          goto TEST;
       }
```

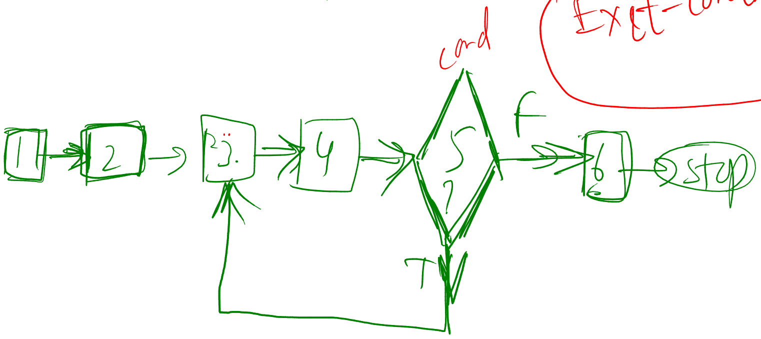
```
}
```

do.. while

Syntax: do{
    Body
}while(bool test condn);

a=~~1~~~~2~~~~3~~~~4~~~~5~~6 =

Exit-controlled loop



```
1) int a=10;
//2) a=1;
   do{
3) pf("In loop a=%d\n",a);
4)   ++a;
5) }while(a<=5);
6) pf("After loop a=%d\n",a);
```

```
main() { int bit, cnt;
         int data;
       pf("Enter int:");
       sf("%d", &data);
bit=31; cnt=0;
  do{   if((data>>bit)&1)
                    cnt++;

        --bit;
  }while(bit>=0);
```

```
bit=31; cnt=0;

TEST:  if ((data>>bit)&1)
                   cnt++;

       --bit;

       if(bit>=0)  goto TEST;
```

W.A.P to Calculate x^y.

I/P	O/P
x: 2 y: 5	32
x: 5 y: 3	125

$x*x*x$

$x=5$, $y=3$

m	x	m=m*x
1	5	m=5
5	5	m=25
25	5	m=125

```
main() { int x, y, m; int cnt;
        Pf("Enter two Ints:");
        Sf("%d %d", &x,&y); // 2, 5
 m=1;  cnt=0;
while (cnt<y)
  {
   m=m*x;        ✓✓
  cnt++;
  }
```

cnt
0✓ 1
1✓
2✓
3✓
4✓
5✗

```
while (y++)
   {
    m *= x; //m = m*x;
   y--;
   }
```

y
5✓
4✓
3✓
2✓
1✓
0
-1

```
int a=10;
// a=1;
while(a<=5)
  {
    Pf("______");
    ++a;
  }
Pf("aft loop\n");
```

W.A.P to verify the Supplied int is a power of 2 or not?

I/P	O/P
32	yes
128	yes
28	NO
10	NO

~~data = 500 0 64~~

data = 20

1
2
4
8
16
32

data = 16

P => 1 → 2 → 4 → 8 → 16

```
int p = 1

while (p < data)
{
  p = p * 2;
}
if (p == data) ________ )
else ________
```

```
if (data & (data - 1) == 0)
      — Yes
else
      — NO
```

```
  010000
& 001111
  000000
```

VPL